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Internal causes and characteristics of high-pour-point oil non-Darcy percolation in porous media under cold damage: An analytical and experimental study

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ABSTRACT

As an essential replacement for conventional crude oils, accurately grasping the flow behavior and threshold pressure characteristics of high-pour-point (high-PPT) oil is pivotal for studying the pressure field and remaining oil distribution in development, and formulating effective strategies. This paper delves into the internal factors contributing to the deterioration of the flow capacity of high-PPT oil, focusing on reservoir and crude oil characteristics, and comprehensively explores its non-Darcy percolation behavior through experiments and analytical methods. The findings highlight that complex reservoir structures, coupled with significant non-Newtonian rheological behavior and notable wax deposition damage, exacerbate the reduction in high-PPT oil flow capacity. Percolation experiments unveil a power-exponential relationship ($G = 5.367 \cdot \lambda_{oil}^{-1.018}$) between the pseudo threshold pressure gradient (PTPG) and oil mobility, constructing a flow boundary chart. Based on this relationship, the critical oil mobility and an ultimate drainage radius formula are derived. Furthermore, leveraging the mechanical equilibrium relationship and fractal theory of porous media, a PTPG prediction model accounting for pore structure characteristics, non-Newtonian characteristics, and wax deposition damage is established. The predicted threshold pressures align well with experimental results, validating the model's efficacy. Notably, PTPG values positively correlate with pore radius fractal dimension, tortuosity fractal dimension, and yield stress of high-PPT oil, while negatively correlating with maximum pore radius and permeability retention degree after wax deposition. These research findings offer valuable insights for designing development schemes for high-PPT oil reservoirs under complex working conditions.

1. Introduction

Crude oil stands as a primary and indispensable energy source worldwide [1,2]. Waxy crudes, constituting approximately 20 % of global petroleum reserves, are increasingly being produced due to the scarcity of more conventional oils [3]. Among these waxy crudes, those with the wax content exceeding 20 % and a pour point surpassing 40 °C are designated as high-pour-point (high-PPT) oils [4–6]. The high wax content leads to relatively elevated wax appearance temperatures (WAT) and pour points (PPT). Traditional water flooding development causes the oil temperature to drop below these critical thresholds. As a result, the originally dissolved wax molecules convert into insoluble

crystals, forming an interlocking gel-like structure [3,7]. This phenomenon not only alters the crude oil's rheology but also leads to the deposition of wax crystals in pipelines and on pore surfaces, impeding the oil flow and resulting in higher pressure drop and pumping costs [8]. Therefore, flow assurance challenges associated with high-PPT oil under low ambient temperature conditions pose significant concerns for oil industries during production, transportation, and storage [9].

Research into the flow characteristics of waxy crudes traces its origins back to 1888, with Zaloziecki's seminal work published in *Angewandte Chemie*, detailing the crystallization of paraffin waxes and its impact on oil viscosity [10]. This longstanding subject has continually progressed over a century, significantly contributing to the evolution of the global energy industry. Padgett *et al.*, [11] conducted pioneering

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Nomenclature			
<i>Symbols</i>		R_e	Ultimate drainage radius
A	Cross-sectional area	R_t	Thickness ratio of the deposition layer
A'	Pre-exponential index	T	Temperature
d	Diameter	T_0	Structural rupture temperature
d_E	Euclidean dimension	v	Mean percolation velocity
D_f	Pore radius fractal dimension	V_c	Cumulative oil flow
D_T	Tortuosity fractal dimension	V_{inj}	Injection volume
E'_a	Pseudo-activation energy	<i>Greek symbols</i>	
G	Pseudo threshold pressure gradient	$\dot{\gamma}$	Shear rate
h	Deposit thickness	η	Viscosity
h_{av}	Average thickness of the deposition layer	λ	Diameter of pore/capillary
K	Permeability	λ_{av}	Average pore diameter
K_e	Effective permeability	λ_{max}	Maximum pore diameter
K_0	Original permeability	λ_{min}	Minimum pore diameter
L	Length	λ_{oil}	Oil mobility
L_t	Actual length of a tortuous capillary	μ	Fluid viscosity
L_0	Straight distance of a tortuous capillary	σ	Wax deposition per unit volume
m	Permeability retention degree	τ	Shear stress
n	Non-Arrhenius index	τ_{av}	Average tortuosity
N	Number of pores/capillaries	τ_0	Yield stress
P_e	Formation pressure	ϕ	Porosity
P_w	Bottom hole pressure	ϕ'	Porosity after wax deposition
ΔP	Pressure difference	ϕ_0	Initial porosity
q	Flow rate in a single capillary	Φ_h	Heat flow
Q	Injection rate or total flow through a cross-section of the core	<i>Abbreviations</i>	
r	Pore radius	DOF	Degree of freedom
r_{av}	Average radius	EOR	Enhanced oil recovery
r'_{av}	Effective pore radius after deposition	PPT	Pour point
r_{max}	Maximum pore radius	PTPG	Pseudo threshold pressure gradient
r_{min}	Minimum pore radius	PV	Pore volume
R	Universal gas constant	WAT	Wax appearance temperature
		WDT	Wax disappearance temperature

research by analyzing the crystallization types of paraffin in various crude oils through wax crystal micrographs. Pandit [12] proposed that natural components in crude oil can lower the PPT and enhance its low-temperature flowability. Simpson [13] first proposed using an oil–water suspension mixture to reduce friction loss in pipelines transporting crude oil with high PPT. Burger *et al.* [14] established the equation of wax deposition thickness as a function of time and distance in long-distance pipelines, accounting for molecular diffusion, shear dispersion, and Brownian motion. Chanda *et al.* [15] proposed for the first time that asphaltenes can influence the flow ability of waxy crudes at low temperatures in conjunction with a polymeric flow improver. Yao *et al.* [16] proposed a synergistic improvement mechanism involving pour point depressants and asphaltenes to improve the flowability of waxy crudes. Wang *et al.* [6] conducted a CT scan study on flow channels and trapped oil distribution in a positive rhythm high-PPT reservoir under various water injection temperatures, proposing three displacement modes and identifying wax deposition, weak water washing degree, and reservoir sedimentary rhythms as key formation mechanisms of trapped oil. Shinde *et al.* [17] performed a numerical simulation on the flow restart operation in a preheated gelled pipeline, demonstrating significant variations in axial velocity profiles for gelled crudes heated above and below the degelation temperature, noting that further heating does not impact the velocity profiles. Despite these advancements, it is notable that these studies have predominantly focused on restarting gelled crudes in pipelines, modeling wax deposition at pipe walls, and examining wax removal forces and efficiency [6]. Comparatively, less attention has been directed towards understanding the percolation behavior of high-PPT oil within reservoirs at low temperatures.

The flow characteristics of crude oil in porous media are intricately influenced and governed by its inherent properties, reservoir characteristics, and interactions [18–21]. In the case of temperature-sensitive high-PPT oil, these factors exert an amplified effect. Existing literature indicates that parameters such as initial oil saturation, temperature, permeability, heterogeneity, and cumulative injection directly influence the wax deposition and flow behavior of high-PPT oil [6,22–26]. From both flow assurance and enhanced oil recovery (EOR) perspectives, it is imperative to comprehend and forecast its percolation behavior within the reservoir.

To achieve the aforementioned objective, this study took the Shen 84-An 12 block of Liaohe Oilfield in China as a case study to elucidate the internal factors contributing to the diminished percolation capacity of high-PPT oil under “cold damage” from two perspectives: reservoir properties and crude oil characteristics. The latter primarily encompassed the rheology of crude oil and its wax deposition capacity. Subsequently, the threshold pressure characteristic was clarified through core percolation experiments. A flow boundary chart was constructed based on the relationship between the pseudo threshold pressure gradient (PTPG) and oil mobility, and an ultimate drainage radius formula for high-PPT oil was derived. Furthermore, considering the fractal characteristics of porous media, non-Newtonian rheological behavior, and wax deposit damage, a PTPG prediction model was established, achieving an absolute error of 9.43 %. The findings contribute to a better understanding and prediction of the non-Darcy percolation characteristics of high-PPT oil in reservoirs.

2. Reservoir characteristics of the Shen 84-An 12 block

Liaohe Oilfield is China's largest production base for high-PPT oil, with high-PPT oil geological reserves of approximate 2.8×10^8 t and an annual production of about 100×10^4 t/a. The Shen 84-An 12 block, situated within Liaohe Oilfield, serves as a prototypical water-injection development block for high-PPT oil. It lies in the central and southern part of the Jing'anpu structural belt in the Damintun sag (Fig. 1), characterized by a faulted nose-like semi-anticline structure [27]. The target layer for development is the third member of the Shahejie Formation, with a burial depth of 1239 – 2375 m. This reservoir features medium porosity and medium-high permeability, with an oil-bearing area of 12.7 km^2 , an effective thickness of 48.7 m, and geological reserves of 6374×10^4 t [28].

The pore structure of porous media plays a decisive role in the porous flow characteristics. Porous media exhibit heterogeneity, with both the granularity of the rock particles and the size of pore throats display sorting features. Pore throats' size, distribution, and geometry are the dominant factors affecting storage and percolation characteristics. The essence of low-velocity nonlinear percolation lies in the gradual participation of pore throats with varying radii, progressing from larger to smaller pore throat radii under the combined influence of the threshold pressure gradient and pore throat radius sorting [21]. Therefore, it is imperative to elucidate the microscopic pore structure characteristics of the study area. The pore structure of 167 natural cores from the Shen 84-An 12 block was examined using a POREMASTER 33 automated mercury porosimeter (Anton Paar, Austria) with a pressure range of 0.2 – 33,000 psia, capable of detecting pore sizes ranging from 1100 to $0.0064 \text{ }\mu\text{m}$. Key parameters for characterizing the pore size of porous media, namely the maximum and average pore radii, are presented in Fig. 2a and 2b. The maximum pore radius is primarily distributed between 6.95 and $26.33 \text{ }\mu\text{m}$, with a mean value of $22.28 \text{ }\mu\text{m}$; meanwhile, the average pore radius is predominantly distributed between 2.45 and $9.91 \text{ }\mu\text{m}$, with a mean value of $6.89 \text{ }\mu\text{m}$. Notably, the maximum and average pore radii exhibit a linear positive correlation with permeability when plotted in a logarithmic coordinate system (with R^2 values of 0.64054 and 0.59025, respectively).

In addition to the aforementioned conventional indicators, the fractal dimension serves as a valuable metric for describing the reservoir characteristics. This is because the microstructure of typical porous media exhibits statistical self-similar fractal characteristics [29]. It can reflect the validity of the space occupied by complex shapes and the measurement of the irregularity of complex shapes. Applied to porous media, fractal dimension quantitatively characterizes the complexity of microstructure such as pore/particle size distribution, pore surface roughness, and streamline bending.

It has been shown that the cumulative size distribution of pores in porous media follows the fractal scaling law:

$$N(L \geq r) = \left(\frac{r_{\max}}{r}\right)^{D_f} \quad (1)$$

where r is the pore radius; $N(L \geq r)$ is the cumulative number of pores with a radius greater than or equal to r ; $r_{\max}$ is the maximum pore radius; and D_f is the pore radius fractal dimension in two dimensions, $1 < D_f < 2$, and is determined by

$$D_f = d_E - \frac{\ln \phi}{\ln(r_{\min}/r_{\max})} \quad (2)$$

where d_E is the Euclidean dimension; ϕ is the porosity; $r_{\min}$ is the minimum pore radius. In general, $r_{\min}/r_{\max} < 10^{-2}$ in porous media.

Capillaries or fluid streamlines in reservoir porous media are not completely straight and exhibit fractal characteristics. The fractal scaling law for the tortuous capillaries in porous media is:

$$L_t = L_0^{\lambda} \lambda^{1-D_T} \quad (3)$$

where λ is the diameter of pore/capillary; L_0 and L_t are the straight distance and actual length of a tortuous capillary, respectively, with $L_t \geq L_0$, and the tortuosity is defined as the ratio of the two; D_T is the fractal dimension for tortuosity, with $1 < D_T < 2$ in two dimensions and $1 < D_T < 3$ in three dimensions.

Considering the average flow characteristics of porous media, i.e., the average tortuosity τ_{av} and the average pore diameter λ_{av} , D_T is defined as:

$$D_T = 1 + \frac{\ln \tau_{av}}{\ln \frac{L_0}{\lambda_{av}}} \quad (4)$$

For fractal porous media, λ_{av} is:

$$\lambda_{av} = \frac{D_f}{D_f - 1} \lambda_{\min} \quad (5)$$

L_0 is defined by Eq. (6):

$$L_0 = \sqrt{\frac{\pi}{4} \frac{D_f}{2 - D_f} \frac{1 - \phi}{\phi} \lambda_{\max}^2} \quad (6)$$

Inserting Eqs. (5) and (6) into Eq. (4) yields:

$$D_T = 1 + \frac{\ln \tau_{av}}{\ln \left(\frac{D_f - 1}{D_f} \frac{\lambda_{\max}}{\lambda_{\min}} \sqrt{\frac{\pi}{4} \frac{D_f}{2 - D_f} \frac{1 - \phi}{\phi}} \right)} \quad (7)$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are the lower and upper limits of the self-similar

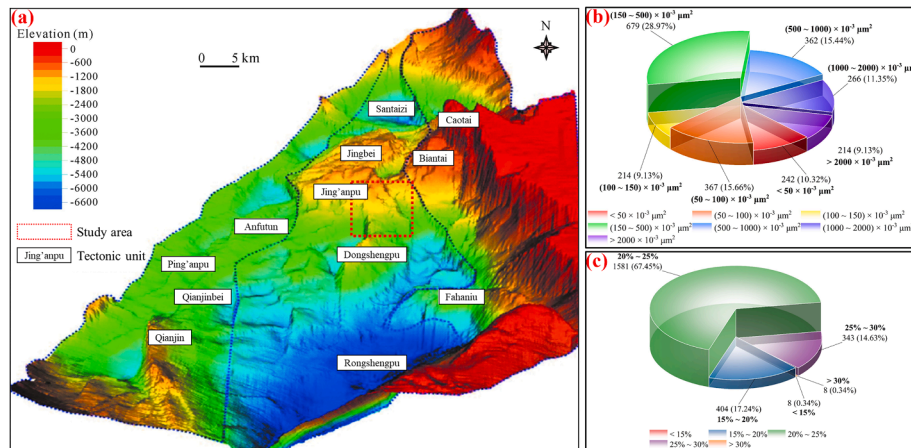


Fig. 1. Overview of the study area. (a) Simplified tectonic map of the Paleogene basement in the Damintun sag;²⁷ (b) permeability, and (c) porosity distribution of the Shen 84-An 12 block.^a ^a These data were calculated from 2344 natural cores.

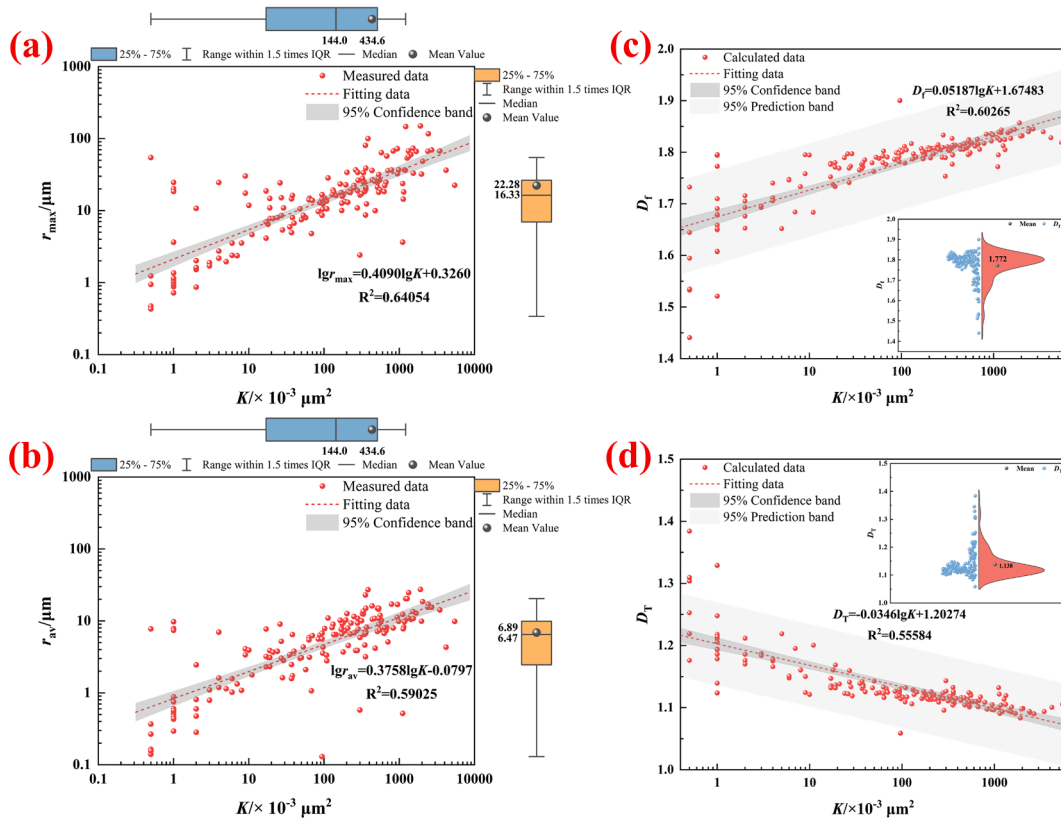


Fig. 2. Pore structure characteristics of the Shen 84-An 12 block. (a) Maximum pore radius, (b) average pore radius, (c) pore fractal dimension, and (d) tortuosity fractal dimension as functions of permeability.

interval of pore size, namely, the minimum and the maximum pore diameters, respectively.

For the average tortuosity of flow paths in porous media, Yu and Li [30] derived a simple geometry model based on the two idealized configurations: some particles are overlapped and the others are not. The tortuosity is obtained by averaging over the two configurations and is expressed as a function of porosity as follows: [29–31]

$$\tau_{\text{av}} = \frac{1}{2} \left[1 + \frac{1}{2} \sqrt{1 - \phi} + \sqrt{1 - \phi} \frac{\sqrt{\left(\frac{1}{\sqrt{1 - \phi}} - 1 \right)^2 + 4}}{1 - \sqrt{1 - \phi}} \right] \quad (8)$$

Therefore, the pore radius fractal dimension D_f and the tortuosity fractal dimension D_T of the Shen 84-An 12 block can be calculated by Eqs. (2), (7), and (8), as depicted in Fig. 2c and 2d. Based on data from the 167 natural cores, D_f decreases and D_T increases with decreasing permeability. D_f ranges from 1.90 to 1.44, with a mean value of 1.772, while D_T ranges from 1.06 to 1.38, with a mean value of 1.138. These findings indicate an uneven distribution of pore sizes within this block, while the tortuosity distribution appears relatively regular. In addition, the dispersion degree of D_f and D_T is significantly affected by permeability. When permeability less than $10 \times 10^{-3} \mu\text{m}^2$, the extensive development of micropore throats leads to a wide range of D_f and D_T . For cores with permeability greater than $10 \times 10^{-3} \mu\text{m}^2$, the fractal characteristics are relatively uniform, with D_f and D_T mainly distributed between 1.75 and 1.86, and 1.08 and 1.17, respectively. The significant variation in reservoir physical properties and the complex pore structure result in deviation from Darcy's law in the percolation behavior of the reservoir, posing potential challenges related to the occurrence of threshold pressure characteristics.

3. Experimental

3.1. Crude oil

The dehydrated crude oil obtained from the production wellhead of the Shen 84-An 12 block is characterized by detailed parameters provided in Table 1. The crude oil predominantly consists of saturated hydrocarbons (>60 %), with a relatively low content of resins and asphaltenes (<25 %), and a wax content of 39.13 %. Fig. 3 illustrates the wax precipitation and melting characteristics determined by TA-DSC25 (TA Instruments, USA) at $5^\circ\text{C}\cdot\text{min}^{-1}$ based on the standard testing procedure. The WAT (66.5°C) and the PPT (47.4°C) demarcate the entire temperature range into three distinct regions. According to the macroscopic flow state of crude oil (the bottle test) and the occurrence state of wax crystals in crude oil (the microscopic photo) in Fig. 3a, these regions are defined as the Newton flow region, wax appearance region, and wax crystal network formation region, respectively. Fig. 3b highlights the wax disappearance temperature (WDT) at 77.1°C , and 10.6°C higher than the WAT, indicating the irreversibility of wax precipitation

Table 1
Physical parameters of oil samples from the Shen 84-An 12 block.

Parameter	Test method	Value
Density	GB/T 1884–2000 (20 °C, 0.1 MPa)	0.865 g·cm ⁻³
Viscosity	GB/T 22235–2008 (70 °C, 0.1 MPa)	16.0 mPa·s
Acid value	GB/T 7304–2014	0.65 mg KOH·g ⁻¹
Relative molecular mass	GB/T 17282–2012	526
Wax content	GB/T 26982–2022	39.13 %
Saturates	NB/SH/T 0509–2010	66.49 %
Aromatics		10.12 %
Resins		4.73 %
Asphaltenes		18.66 %

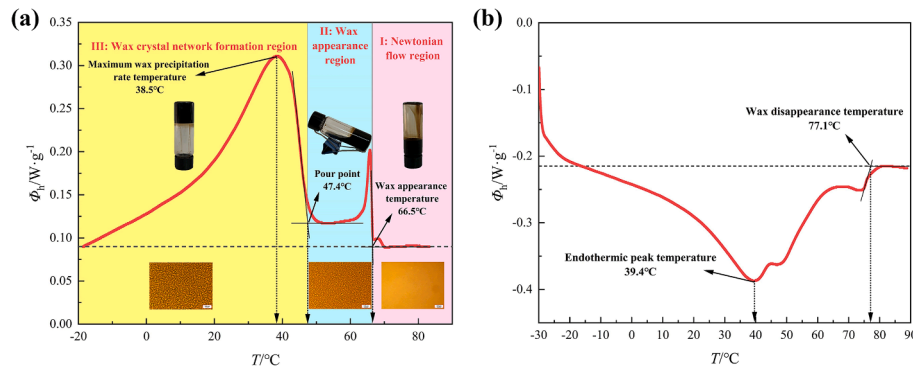


Fig. 3. Heat flow curves of high-PPT oil. (a) Cooling and (b) heating processes.

in high-PPT oil. To ensure enhanced reproducibility of the experimental data, the oil samples underwent pre-treatment to eliminate the “memory” effect associated with shear and thermal history [32].

3.2. Rheological test of high-PPT oil

The viscosity-temperature relationship during the cooling process (from 90 to 25 °C, with a cooling rate of 0.02 °C·s⁻¹) at different shear rates (7.34 to 200 s⁻¹) and the stress-shear rate relationship with a linear increase in shear rate from 0 to 100 s⁻¹ at various temperatures were elucidated using the CC27 concentric cylinder system of Physica MCR301 (Anton Paar, Austria). To ensure accuracy, the reported results represent the average of three measurements.

3.3. Core wax deposition experiment of high-PPT oil

According to the reservoir characteristics of the Shen 84-An 12 block, one-dimensional sandstone cores (diameter $d = 2.5$ cm, length $L = 10$ cm) with varying gas permeabilities (about 150, 400, and 800 × 10⁻³ μm²) were designed based on the artificial core fabrication technology of full pore space distribution [33,34]. These cores were used for wax deposition experiments, with pore throat structure parameters determined via mercury injection experiments. Fig. 4 depicts the experimental apparatuses, and the specific experimental procedure was as follows: (1) vacuum the core and saturate it with kerosene; (2) calculate the core porosity based on the saturated kerosene content; (3) replace the kerosene by crude oil at 0.1 mL·min⁻¹ in an 80 °C thermostat (higher than the WDT) until no kerosene was produced at the outlet and the injection pressure stabilized; (4) calculate the initial oil permeability according to Darcy’s law; (5) age the core for 24 h at the experimental temperature; (6) inject a total of 2 times the pore volume (PV) of crude oil at 0.1 mL·min⁻¹ and record the injection pressure every 0.1 PV; (7) calculate the effective permeability after wax deposition and establish

the wax deposition model to describe the wax deposition law. Each experiment was conducted three times for accuracy of results.

3.4. Pseudo threshold pressure gradient experiment of high-PPT oil

The cores and the experimental apparatuses for this part were consistent with those described in Sec. 3.3. The experimental process proceeded as follows: first, prepare the core for testing and calculate the porosity and oil permeability following steps (1) to (5) outlined in Sec. 3.3. Next, inject crude oil incrementally from low to high flow rates, adjusting the rate when the injection pressure stabilized. Finally, plot and linearly fit the relationship between injection rate and pressure gradient. The intercept of the fitting line on the pressure gradient coordinate axis represents the PTPG value. All measurements were conducted thrice to validate reproducibility.

4. Results and discussion

4.1. Non-Newtonian rheological behavior of high-PPT oil

The particularity of high-PPT oil is mainly reflected in its extremely sensitive nature to temperature. It exhibits a temperature-dependent transition from liquid-like to solid-like upon cooling or vice versa, which leads to significant rheological changes in these “thermos-reversible gels” [17]. At different temperatures, wax undergoes distinct changes in its occurrence state, leading to variations in the phase state and rheological properties of the high-PPT oil. When the temperature exceeds WAT, the crude behaves as a Newtonian fluid, with viscosity solely dependent on temperature changes. Below the WAT, wax crystals precipitate and gradually form a three-dimensional network structure, transforming the crude into a non-Newtonian fluid whose viscosity varies with shear rate $\dot{\gamma}$ and rises rapidly as the temperature decreases. Furthermore, when the temperature drops below PPT, the wax crystal

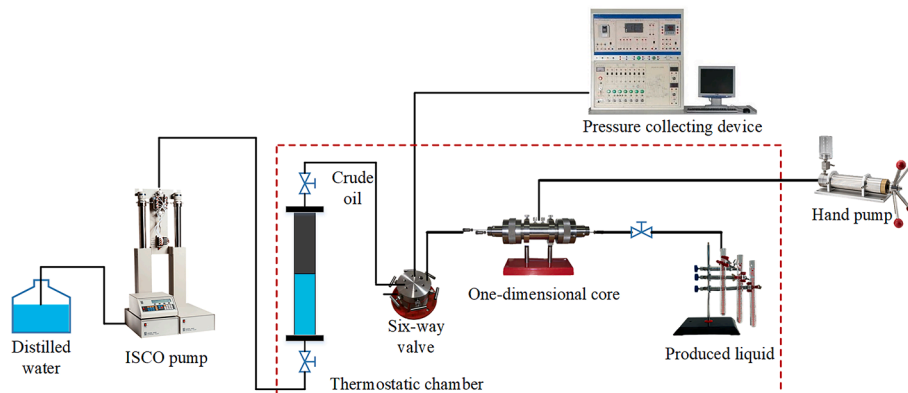


Fig. 4. Schematic diagram of measuring the wax deposition damage and the pseudo threshold pressure gradient.

network structure is completely formed, the crude loses its fluidity, and the viscosity jumps further. Accordingly, the viscosity-temperature ($\eta-T$) curve of high-PPT oil (Fig. 5a) exhibits a unique three-segment broken line in the semi-logarithmic coordinate system before reaching the structural rupture temperature T_0 (about 305 K for the high-PPT oil of the Shen 84-An 12 block). This indicates a sharp decline in the flow capacity of high-PPT oil with decreasing temperature and a deviation of the $\eta-T$ relationship from the conventional Arrhenius model, which typically appears as a straight line in the semi-logarithmic coordinate system and is only applicable within a relatively small temperature and viscosity range [35,36]. Considering the structural rupture temperature and the non-Arrhenius temperature dependence of high-PPT oil, T_0 and the non-Arrhenius index n (where $n \leq 1$, and smaller n values indicate less temperature dependence of viscosity) are introduced to modify the Arrhenius model, resulting in the modified Arrhenius model, as shown in Eq. (9). When $T_0 = 0$, $n = 1$, it reverts to the conventional Arrhenius form.

$$\eta = A' e^{\frac{E'_a}{R(T-T_0)^n}} \quad (9)$$

where, A' is a pre-exponential term linked to the entropy of activation, mPa·s; E'_a represents a pseudo-activation energy related to the potential energy barriers obstructing the structural rearrangement, J·mol⁻¹; R is the universal gas constant with a value of 8.314 J·mol⁻¹·K⁻¹; and T is the absolute temperature, K.

As shown in Fig. 5a, the modified Arrhenius model was utilized to fit the $\eta-T$ relationship at different shear rates, achieving a goodness of fit greater than 0.98, indicative of excellent fitting and prediction capabilities. Fitting parameters $\ln A'$, E'_a , and n exhibited polynomial, power function, and linear decreasing forms, respectively, with increasing shear rate $\dot{\gamma}$ (Fig. 5b, with accuracies exceeding 0.99). On the one hand, the increase in shear rate helps overcome intermolecular interactions, enhances molecular motion, aligns molecules along the flow direction,

and reduces the degree of freedom (DOF) of the molecules in this direction, thus resulting in a reduction of the system's entropy of activation and counterbalancing the variation of viscosity with temperature to a certain extent; on the other hand, the change in shear rate alters the molecular arrangement in the system, modifying the height and “shape” of the energy barrier and facilitating it easier to overcome.

Introducing the equations in Fig. 5b into Eq. (9) yields the relationship among viscosity, temperature, and shear rate for high-PPT oil in the Shen 84-An 12 block, as shown in Eq. (10). Leveraging this derivation, the viscosity heatmap of high-PPT oil (Fig. 5c) is constructed, facilitating the swift determination of viscosity values under specific working conditions to satisfy the requirement of oilfields.

$$\ln \eta = -0.0005\dot{\gamma}^2 + 0.017\dot{\gamma} - 5.8835 + \frac{502.5692\dot{\gamma}^{-0.15523}}{R} \cdot \frac{1}{(T - T_0)^{-0.00175\dot{\gamma} + 0.4827}} \quad (10)$$

The deterioration of the flow ability of high-PPT oil due to the decrease in temperature not only increases its viscosity, but also significantly elevates its yield stress τ_0 . Fig. 6 shows that the slope of the $\tau-\dot{\gamma}$ relationship curve gradually increases with the temperature decrease, and the intercept on the ordinate axis increases gradually. This trend signifies that the reduction in temperature imparts pronounced non-Newtonian characteristics to high-PPT oil, resulting in substantial increases in viscosity and yield stress. When the temperature surpasses the WAT, the crude oil is in a Newtonian fluid state, and the $\tau-\dot{\gamma}$ curve forms a straight line passing through the origin with negligible yield stress (0–0.0006 Pa). However, below the WAT, with the decrease in temperature, the microstructure of high-PPT oil changes, with needle-like wax crystals gradually accumulating and interweaving to form a more intricate network structure. When shear force is applied, it becomes necessary to overcome the mutual attraction between the wax crystals and the strength of the formed network structure to initiate oil

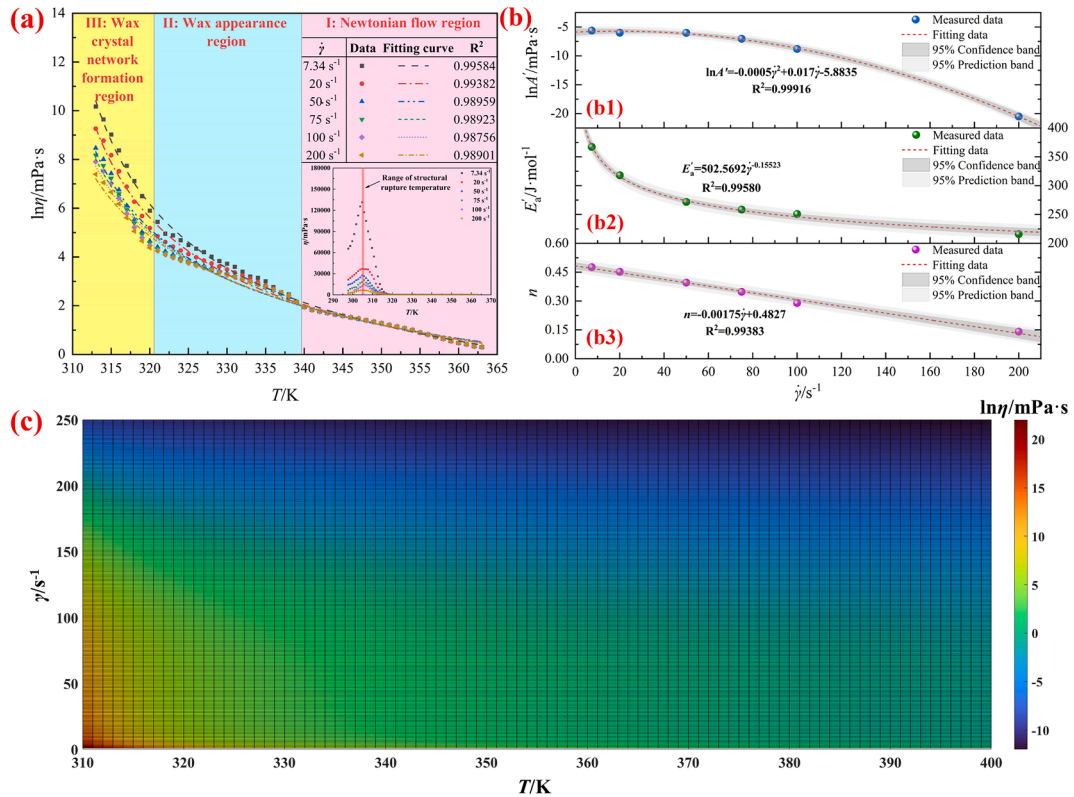


Fig. 5. Fitting of viscosity-temperature relationship of high-PPT oil by the modified Arrhenius model: (a) original data and fitting results, (b) fitting parameters as functions of shear rate, and (c) viscosity heatmap.

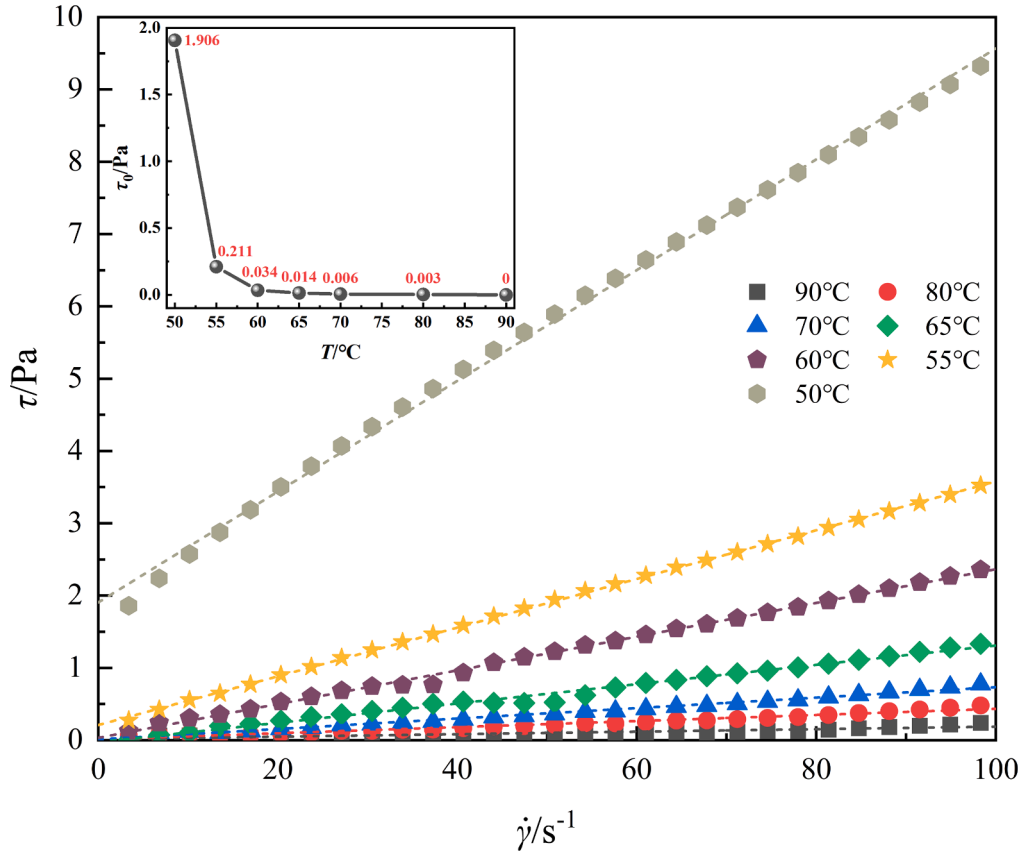


Fig. 6. Stress-shear rate relationship of high-PPT oil at various temperatures.

flow, resulting in a sharp increase in the yield stress with decreasing temperature.

Overall, the increase in viscosity and yield stress caused by the deterioration of the rheological properties under “cold damage” is one of the internal causes of the threshold pressure characteristics of high-PPT oil in reservoir pores.

4.2. Wax deposition law of high-PPT oil in porous media

Wax deposition in the reservoir significantly impacts well productivity in waxy crude oilfields. In the original reservoir environment, the wax is dissolved in the crude oil in molecular form. However, wax molecules crystallize and deposit on the pore surface when the temperature drops to the WAT during cold-water injection. As deposition progresses, the wax crystals gradually form a three-dimensional network structure, trapping a large amount of crude oil and forming a wax deposition layer. This process inevitably leads to a reduction in reservoir permeability and porosity, ultimately diminishing oil recovery. Given these implications, it is imperative to elucidate the wax deposition behavior of high-PPT oil in porous media. Understanding this phenomenon is crucial for optimizing production strategies and mitigating potential productivity losses in waxy crude oil reservoirs.

4.2.1. Establishment of wax deposition model in porous media

For high-PPT oil percolation, the effective permeability after wax deposition K_e is a multi-variable function ($K_e = f(K_0, T, V_c)$) based on independent variables such as original permeability K_0 , temperature T , and cumulative oil flow V_c . To facilitate the study, the core is simplified into a series of capillary bundles with the average radius of r_{av} , and the deposited wax is equivalent to a uniformly coated ring on the inner wall of the capillary [25,37]. The thickness of the deposition layer is denoted as h , and the effective pore radius after deposition is r'_{av} . Suppose that the

core to be studied is a columnar core with a length of L , a cross-sectional area of A , and the number of capillaries per unit area is N with an average tortuosity of τ_{av} . Under the condition that the real core and the imaginary core have the same apparent size, the same fluid properties, and the same pressure difference, the permeability K can be obtained from the Poiseuille equation and Darcy's law as follow:

$$K = N \frac{\pi r_{av}^4}{8 \tau_{av}} \quad (11)$$

According to the definition, the porosity of the model can be obtained:

$$\phi = \frac{\text{Pore volume}}{\text{Rock bulk volume}} = \frac{AN\pi r_{av}^2 L \tau_{av}}{AL} = N\pi r_{av}^2 \tau_{av} \quad (12)$$

Assuming that the initial porosity is ϕ_0 , the porosity after wax deposition is ϕ' , the wax deposition per unit volume is σ , and the permeability retention after wax deposition $m = K_e/K_0$, then:

Considering that wax deposition does not affect the tortuosity of capillaries, the average thickness of the deposition layer is:

$$\frac{K_e}{K_0} = \frac{(r'_{av})^4}{r_{av}^4} = \frac{(r_{av} - h_{av})^4}{r_{av}^4} \Rightarrow h_{av} = r_{av} \left(1 - \sqrt[4]{\frac{K_e}{K_0}}\right) = r_{av} (1 - \sqrt[4]{m}) \quad (13)$$

Similarly, the deposit thickness in any pore is:

$$h = r \left(1 - \sqrt[4]{m}\right) \quad (14)$$

The thickness ratio of the deposition layer is:

$$R_t = \frac{h_{av}}{r_{av}} \times 100\% = \left(1 - \sqrt[4]{m}\right) \times 100\% \quad (15)$$

The porosity after wax deposition is:

$$\frac{\phi'}{\phi_0} = \frac{N\pi(r'_{av})^2 \tau_{av}}{N\pi r_{av}^2 \tau_{av}} = \frac{(r'_{av})^2}{r_{av}^2} = \sqrt{\frac{K_e}{K_0}} \Rightarrow \phi' = \phi_0 \sqrt{\frac{K_e}{K_0}} = \phi_0 \sqrt{m} \quad (16)$$

The wax deposition per unit volume is:

$$\sigma = \frac{AL(\phi_0 - \phi')}{AL} = \phi_0 - \phi' = \phi_0 \left(1 - \sqrt{\frac{K_e}{K_0}}\right) = \phi_0(1 - \sqrt{m}) \quad (17)$$

These equations provide a theoretical framework for understanding wax deposition behavior in porous media, enabling the assessment of reservoir performance and the development of effective mitigation strategies.

4.2.2. Quantitative evaluation of wax deposition damage to the reservoir

Fig. 7 illustrates the relationship between the permeability retention m after wax deposition for three kinds of cores (initial gas permeabilities of 800, 400, and $150 \times 10^{-3} \mu\text{m}^2$, with corresponding initial oil permeabilities of 230, 120, and $50 \times 10^{-3} \mu\text{m}^2$, respectively) and the injection volume V_{inj} of high-PPT oil. As V_{inj} increases, m exhibits a rapid decrease followed by a gradual slowing of the decline. Eventually, m stabilizes, with lower experimental temperatures associated with smaller m values at each stage. Moreover, cores with lower initial permeabilities experience more significant decreases in effective permeability for a fixed temperature and injection volume. Notably, the injection volume corresponding to wax deposition dynamic equilibrium (highlighted by the red pentagram) decreases with decreasing temperature and permeability.

The above phenomena can be explained by Fig. 8. When the temperature falls below the WAT, wax crystals initially disperse uniformly and randomly within the bulk oil (Stage I). As oil flow continues and velocity gradients form within the capillary, wax content in the bulk oil gradually increases. Concurrently, wax crystals undergo radial diffusion influenced by various factors such as molecular diffusion, Brownian motion, Soret diffusion, and gravity settlement, leading to their accumulation on the pore surface (Stage II) [38,39]. This accumulation accelerates the formation of the three-dimensional network structure, enveloping a significant volume of oil and increasing resistance to fluid flow. Consequently, the layer near the pore surface develops yield stress, which progressively rises with the accumulation of wax crystals. Despite this, the weak flow layer persists in motion as long as its yield stress remains lower than the shear stress imposed by the fluid flow (Stage III) [26]. However, once the yield stress surpasses the local shear stress, the weak flow layer ceases to flow and transforms into a deposit layer, where wax deposition and shear stripping reach dynamic equilibrium (Stage IV). Lower ambient temperatures increase wax crystal precipitation and promote the formation of a more robust network structure. Similarly, lower initial core permeability, indicating smaller average pore radii, facilitates the accumulation of wax crystals on the pore

surface and the formation of a thicker and more robust deposit layer. Consequently, lower temperatures and initial permeabilities exacerbate the “cold damage” induced by wax deposition.

After clarifying the permeability decrease caused by wax deposition, the m value of wax deposition equilibrium in Fig. 7 is incorporated into Eqs. (13)–(17) to quantitatively characterize the “cold damage” degree. Fig. 9 presents the comparison between the wax deposition thickness and the average pore radius, alongside the proportion of wax deposition layer thickness. Meanwhile, Fig. 10 compares the porosity before and after wax deposition and presents the wax deposition per unit volume. As illustrated in Fig. 9, with the decrease in temperature and permeability, the thickness of the wax deposition layer increases significantly. For cores initially having $K_0 = 800 \times 10^{-3} \mu\text{m}^2$, h_{av} ranges from 0.38 to $1.26 \mu\text{m}$ across temperatures, with corresponding thickness ratios R_t of 3.68 % to 12.23 %; when K_0 decreases to $150 \times 10^{-3} \mu\text{m}^2$, h_{av} increases to 1.13 to $1.97 \mu\text{m}$, with R_t exceeding 20 % and reaching a maximum of 35.25 %. These findings align closely with results reported by Wang et al., who conducted wax deposition experiments using live crude at 55°C and 12 MPa [24]. As depicted in Fig. 10, the initial porosity of the cores is about 26 %, experiencing varying degrees of loss post-experimentation. For $K_0 = 800 \times 10^{-3} \mu\text{m}^2$, the porosity after wax deposition ranges from 23.69 % to 19.43 %, indicating an insignificant reduction. However, at lower temperatures and permeabilities, the effective porosity of the core suffers serious loss. For instance, at $K_0 = 150 \times 10^{-3} \mu\text{m}^2$ and $T = 50^\circ\text{C}$, ϕ' drops to 10.91 %, marking a reduction of over 50 %. Correspondingly, the wax deposition per unit volume σ exhibits a strong negative correlation with temperature and permeability. As K_0 and T decrease from $800 \times 10^{-3} \mu\text{m}^2$ and 65°C to $150 \times 10^{-3} \mu\text{m}^2$ and 50°C , respectively, σ jumps from 0.017 to $0.139 \text{ cm}^3 \cdot \text{cm}^{-3}$ representing a more than sevenfold increase.

In summary, with the increase of crude oil injection volume and the decrease of permeability and temperature, the “cold damage” induced by wax deposition escalates. This results in a reduced effective percolation area within the reservoir, leading to an increased injection-production pressure difference. Moreover, the effective porosity of the reservoir diminishes, contributing to decreased production efficiency.

4.3. Threshold pressure characteristic of high-PPT oil

High-PPT oil, being temperature-sensitive, translates into non-Newtonian fluid below the WAT, altering its percolation characteristics. Additionally, the deposited wax obstructs the reservoir pores, diminishing effective permeability, increasing flow resistance, and exacerbating reservoir heterogeneity. Consequently, water injection pressure rises, significantly impacting development efficiency. Precisely understanding the threshold pressure characteristic of high-PPT oil is crucial for devising technical strategies such as development bound-

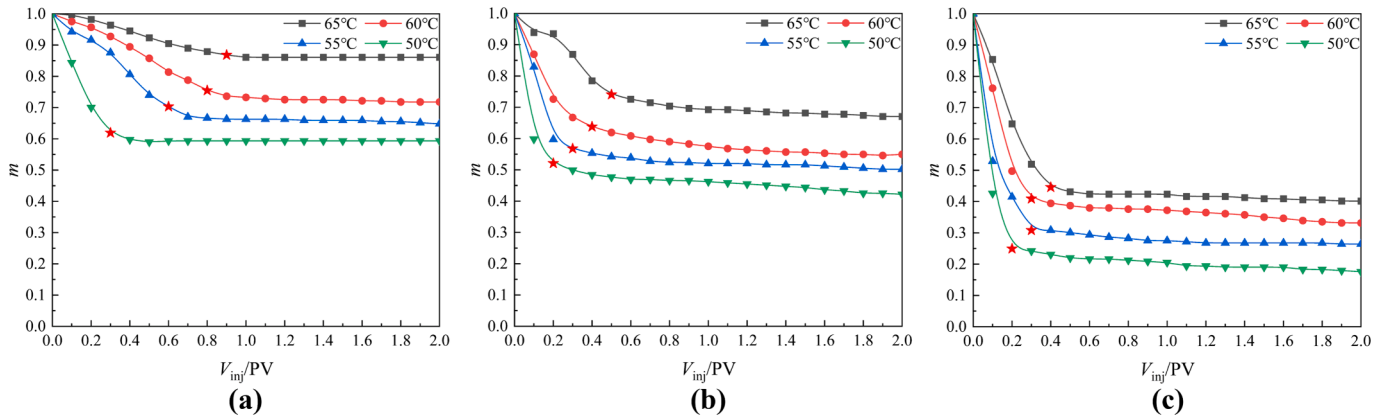


Fig. 7. Relationship between the permeability retention degree and injection volume. Initial gas permeabilities about: (a) $800 \times 10^{-3} \mu\text{m}^2$, (b) $400 \times 10^{-3} \mu\text{m}^2$, and (c) $150 \times 10^{-3} \mu\text{m}^2$.

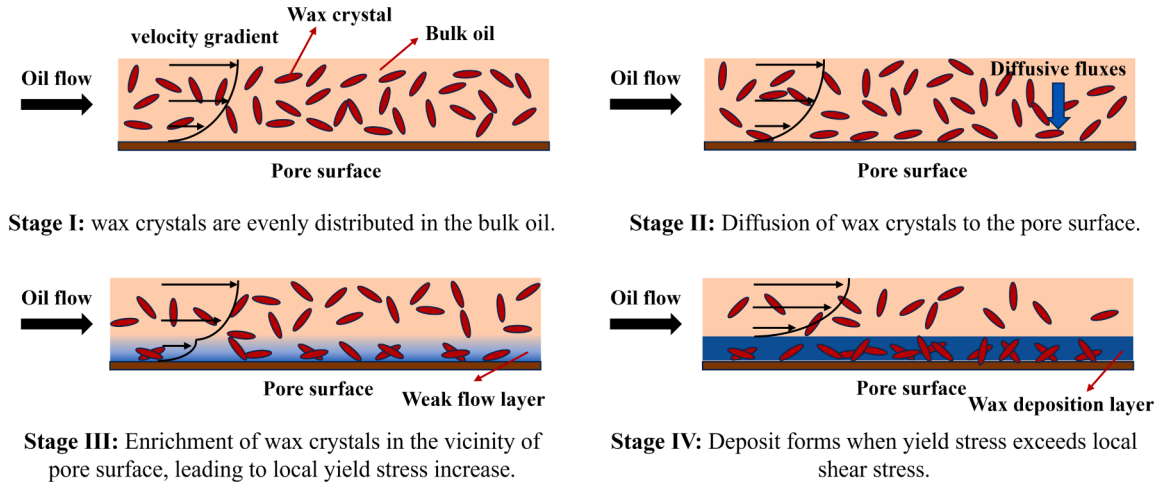


Fig. 8. Illustration of the deposit formation in the porous media from the rheological perspective.

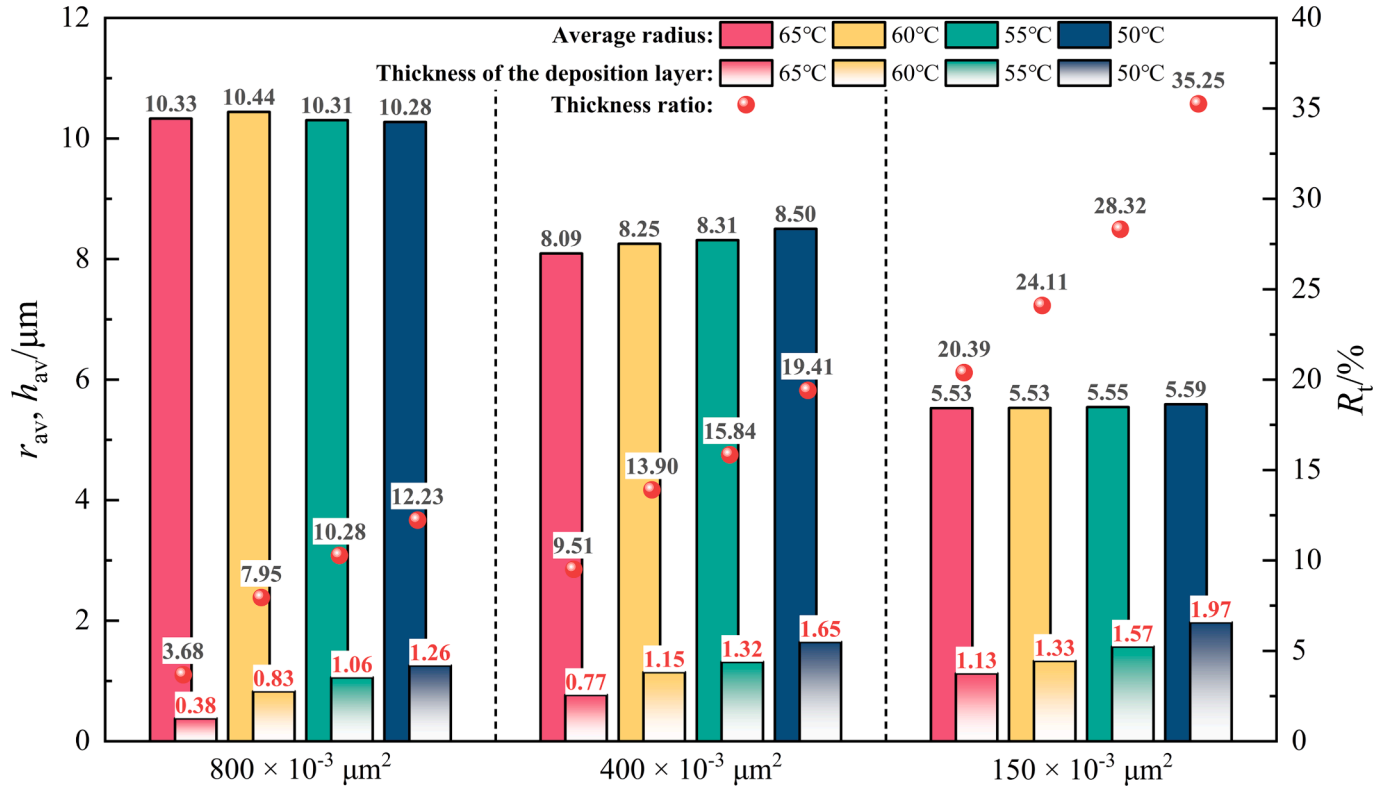


Fig. 9. Comparison of wax deposition thickness and average pore radius under different temperature and permeability conditions.

aries, well pattern and spacing design, and optimization of development schemes for such reservoirs [40]. The percolation behavior of high-PPT oil in porous media can be characterized by the modified Darcy's law, expressed as:

$$v = 0 \quad \left(\text{when } \frac{\Delta P}{L} < G \right) \quad (18a)$$

$$v = \frac{K}{\mu} \left(\frac{\Delta P}{L} - G \right) \quad \left(\text{when } \frac{\Delta P}{L} \geq G \right) \quad (18b)$$

where v is the mean percolation velocity; K is the permeability; ΔP is the pressure difference; L is the core length; μ is the fluid viscosity; and G is the pseudo threshold pressure gradient.

As shown in Fig. 11, the relationship between the pressure gradient

$\Delta P/L$ of the steady-state flow of high-PPT oil in the core and the injection rate Q (proportional to v) under varying permeability and temperature conditions is presented, with the linear section of the scatter diagram fitted. The slope of the fitting line gradually diminishes with decreasing temperature and permeability, while the intercept on the abscissa axis increases, indicating higher PTPG values and reduced percolation capacity for high-PPT oil. Above the WAT, the PTPG is negligible, measured at 0.001, 0.012, and 0.090 $\text{MPa}\cdot\text{m}^{-1}$ (70 $^{\circ}\text{C}$), respectively. This is primarily due to the absence of wax crystal precipitation in high-PPT oil above the WAT, with minimal viscoelasticity, resembling a Newtonian fluid and avoiding wax deposition damage [4]. Consequently, it adheres to traditional Darcy's law. However, as temperature decreases, oil viscosity rises, and the number and size of the wax crystals also increase significantly, thereby enhancing the

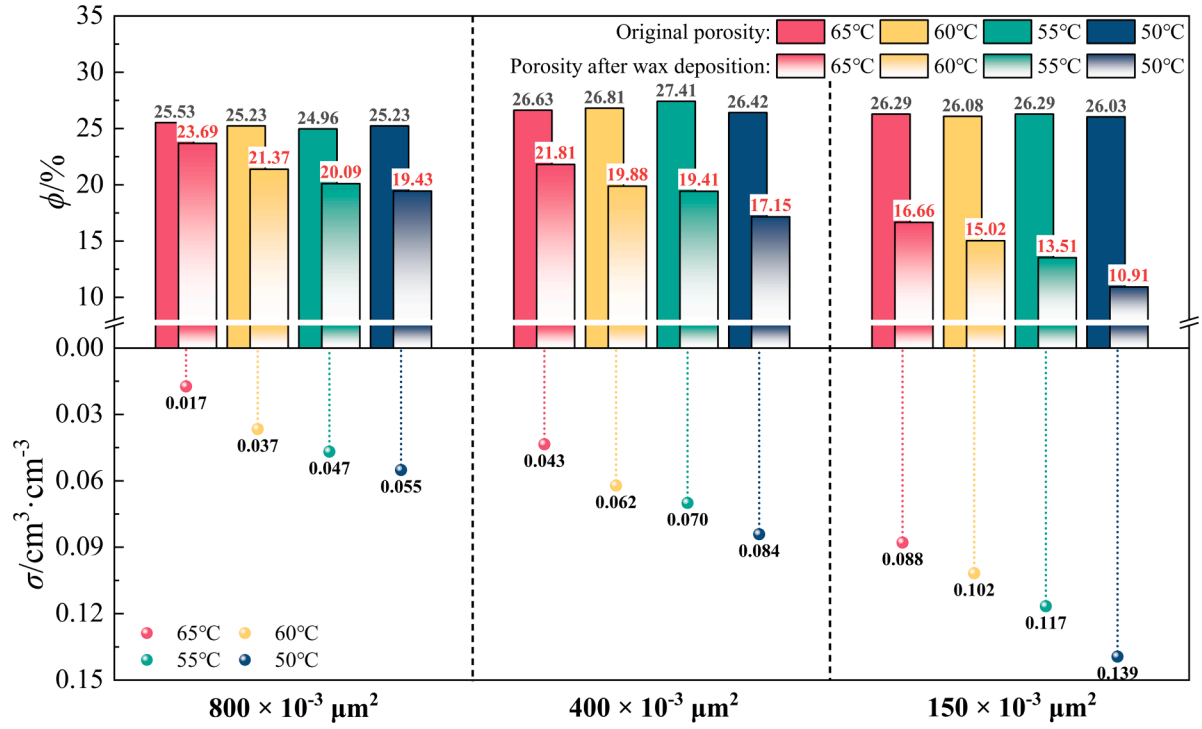


Fig. 10. Comparison of porosity before and after wax deposition and wax deposition per unit volume under different temperature and permeability conditions.

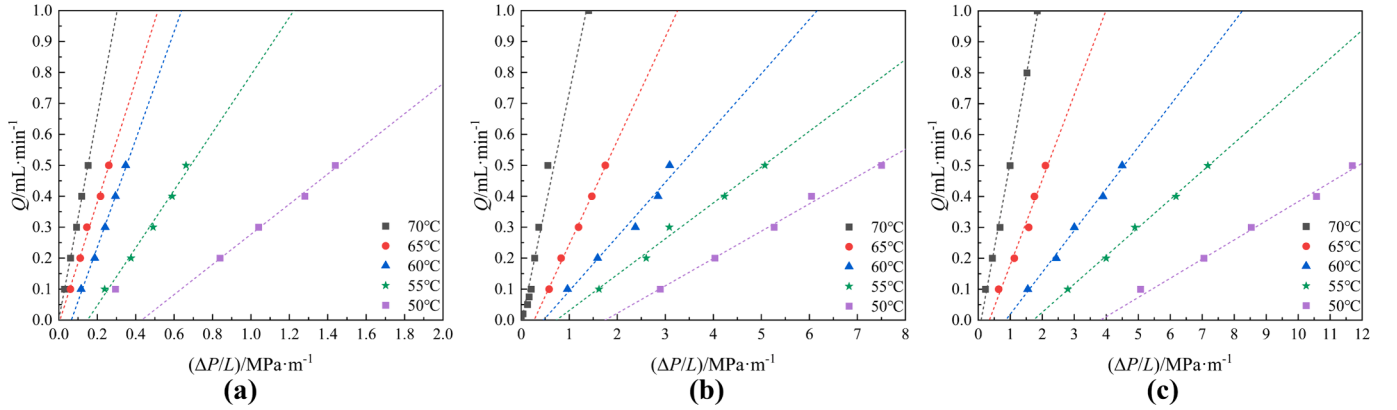


Fig. 11. Percolation curves of high-PPT oil in cores with various gas permeabilities: (a) $800 \times 10^{-3} \mu\text{m}^2$, (b) $400 \times 10^{-3} \mu\text{m}^2$, and (c) $150 \times 10^{-3} \mu\text{m}^2$.

viscoelasticity of high-PPT oil and the wax deposition thickness along with its yield strength. Moreover, reduced permeability signifies increased complexity in pore structure, exacerbating wax deposition damage. Consequently, under low-temperature and low-permeability conditions, the PTPG of high-PPT oil becomes prominent. At 50°C , the PTPG in the three cores measures 0.431 , 1.749 , and $3.818 \text{ MPa}\cdot\text{m}^{-1}$, respectively. To visually illustrate the influence of permeability and temperature on the PTPG, a 3D diagram is presented in Fig. 12.

As previously discussed, the threshold pressure characteristic comprehensively embodies reservoir and fluid properties. Oil mobility λ_{oil} , the ratio of reservoir permeability to oil viscosity, serves as a metric for crude oil's flow capacity in porous media. Hence, the $G-\lambda_{\text{oil}}$ relationship can be established by integrating information from Figs. 5 and 12. As depicted in Fig. 13a, a smaller λ_{oil} necessitates a higher G . The $G-\lambda_{\text{oil}}$ relationship conforms to a remarkable power-law form [Eq. (19)], with a goodness of fit of 0.9790 . The fitting curve highlights a critical oil mobility of $2.3 \times 10^{-3} \mu\text{m}^2 \cdot (\text{mPa}\cdot\text{s})^{-1}$. When λ_{oil} falls below this threshold, G increases sharply, indicating the difficulty and limited exploitation potential of high-PPT oil in the Shen 84-An 12 block within

this range.

$$G = 5.367 \cdot \lambda_{\text{oil}}^{-1.018} \quad (19)$$

In the logarithmic coordinate system, the $G-\lambda_{\text{oil}}$ relationship exhibits a linear form, dividing the plane into two regions (Fig. 13b). The upper section represents the flowable region, where the pressure gradient surpasses the PTPG, allowing crude oil to exhibit pseudo-linear flow under the given pressure difference. In contrast, the lower section denotes the non-flowable zone, where the crude oil either does not flow or only exhibits low-speed nonlinear flow. Therefore, Fig. 13 serves as a quick reference to determine the underground flow state of crude oil.

Furthermore, the relationship between the ultimate drainage radius (R_e) and oil mobility, coupled with the production pressure difference in the Shen 84-An 12 block can be derived by substituting Eq. (19) into the ultimate drainage radius formula [Eq. (20)], as shown in Eq. (21): [41,42]

$$R_e = \frac{P_e - P_w}{\Delta P/L} \quad (20)$$

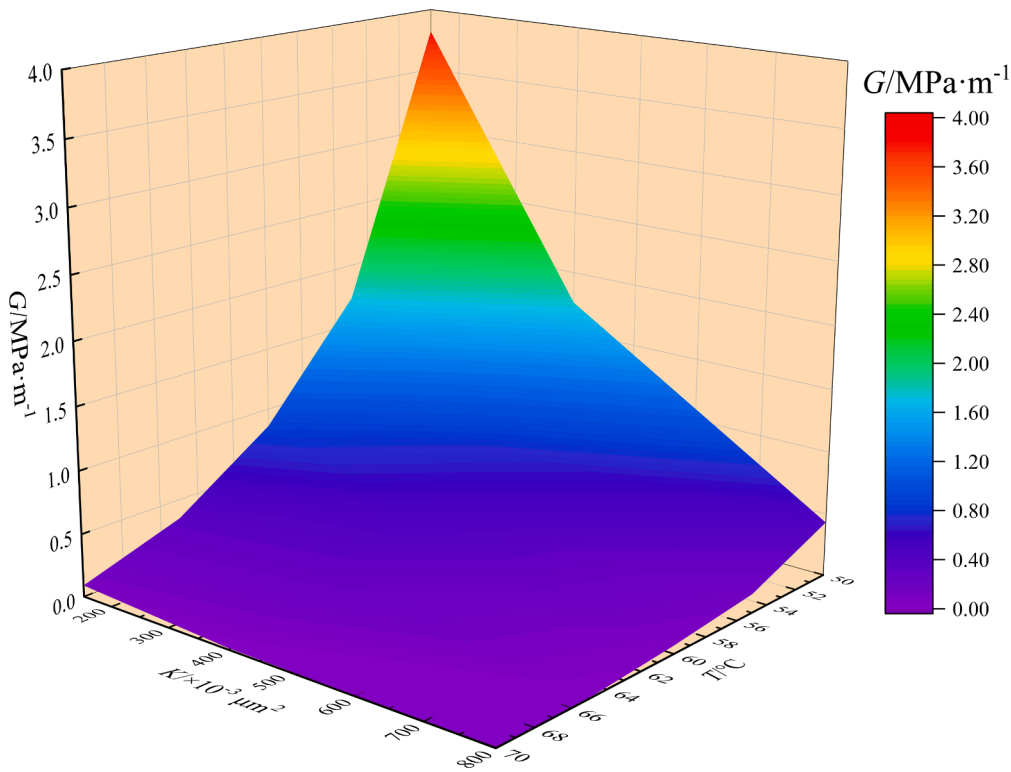
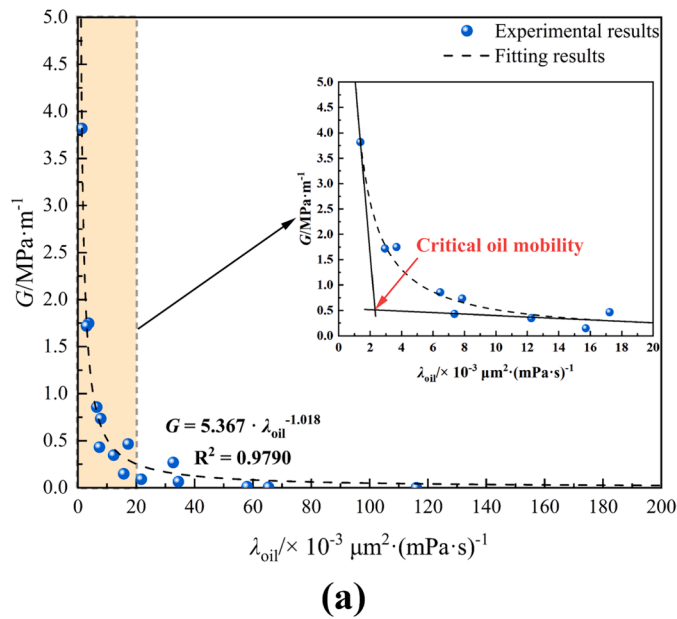
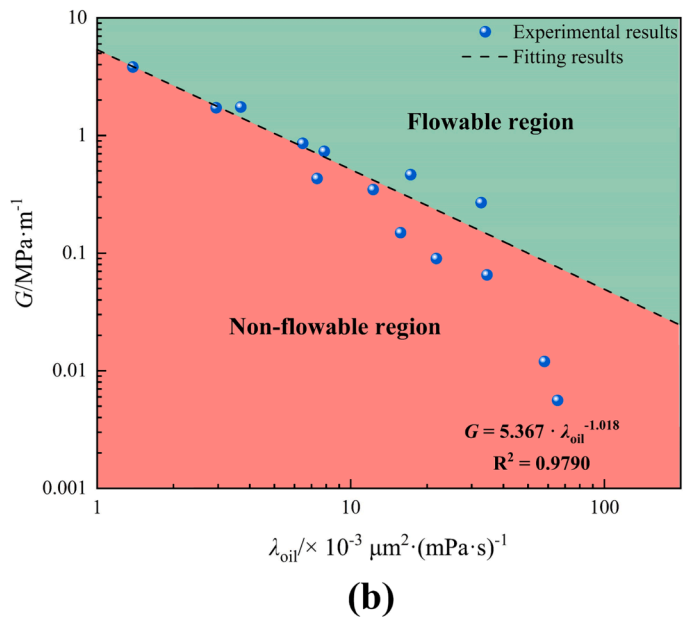


Fig. 12. Pseudo threshold pressure gradient under different temperature and permeability conditions.



(a)



(b)

Fig. 13. Flow boundary charts of high-PPT oil in forms of (a) natural coordinate axis and (b) logarithmic coordinate axis.

$$R_e = 0.186 \bullet (P_e - P_w) \bullet \lambda_{oil}^{1.018} \quad (21)$$

where P_e denotes the formation pressure, and P_w represents the bottom hole pressure.

Using Eq. (21), with known reservoir permeability and oil viscosity, one can predict R_e under specific production pressure differences, guiding the determination of well spacing. For instance, the average permeability of the Shen 84-An 12 block is $413 \times 10^{-3} \mu m^2$, the formation pressure is 19.3 MPa, and the oil viscosities are 7.78, 12.24, and

24.52 mPa·s at 70, 65, and 60 °C, respectively. Assuming the bottom hole pressure is 5 MPa, R_e would approximate 152 m, 96 m, and 47 m at these respective temperatures. This highlights that after encountering “cold damage,” the development potential of high-PPT oil diminishes swiftly. Consequently, well-controlled reserves, recoverable reserves, and cumulative oil production from wells will also decrease significantly. Mitigating measures such as well pattern densification are crucial to establishing favorable injection-production relationships and achieving efficient development.

5. Prediction model for pseudo threshold pressure gradient

While laboratory experiments are effective in determining threshold pressure, they are both expensive and time-consuming. As an alternative, analytical methods have also been developed to study threshold pressure, which are less data-intensive than numerical methods and more economical than experimental approaches. As detailed in Sec. 2, the Shen 84-An 12 block exhibits complex pore structure characteristics, whose influence on threshold pressure cannot be overlooked. Additionally, the rheological deterioration of high-PPT oil due to the “cold damage” increases flow resistance in porous media. Moreover, as alluded in Sec. 4.2 and in existing literature, wax deposition in porous media plays a crucial role in evaluating properties such as porosity, permeability, and threshold pressure in high-PPT oil reservoirs [6,24,43]. Previous studies have not comprehensively consider the influence of these factors, particularly wax deposition [18,44,45]. Thus, it is imperative to comprehensively account for the influence of pore structure characteristics, non-Newtonian characteristics, and wax deposition damage when establishing the PTPG prediction model. Hereby, the model can be derived under the following assumptions:

- (1) The pore structure has self-similar characteristics, allowing the reservoir core to be modeled as an unequal-diameter bending capillary bundle that adheres to fractal theory;
- (2) The rheological characteristic of high-PPT oil obeys the Bingham fluid model equation;
- (3) The flow process is isothermal and steady-state, and the pressure drop is uniformly distributed in the pore space during the flow process;
- (4) When wax deposition occurs, the wax precipitated uniformly coating the pore surface in film form.

The force balance of high-PPT oil in a single curved capillary under flow conditions is investigated. According to the mechanical equilibrium, we have:

$$\Delta P \cdot \pi r^2 - \left(\tau_0 + \mu \frac{dv}{dr} \right) \cdot 2\pi r L_t = 0 \quad (22)$$

Integrating Eq. (3) into Eq. (22), the fluid velocity is obtained as:

$$v = \frac{\Delta P \cdot 2^{D_T-2} \cdot r^{D_T+1}}{(D_T+1)L_0^{\frac{D_T}{\mu}} \mu} - \frac{\tau_0 r}{\mu} \quad (23)$$

Considering the reduction of effective percolation area caused by wax deposition, the flow rate of high-PPT oil in a single capillary is:

$$\begin{aligned} q &= \int_0^{r-h} v \cdot 2\pi r dr \\ &= \int_0^{r_{\max}^{\sqrt{m}}} \frac{\Delta P \cdot 2^{D_T-1} \cdot \pi r^{D_T+2} \cdot L_0^{-D_T}}{\mu (D_T+1)} - \frac{2\pi \tau_0 r^2}{\mu} dr \\ &= \frac{\Delta P \cdot 2^{D_T-1} \cdot \pi \cdot L_0^{-D_T} \cdot m^{\frac{D_T+3}{4}}}{\mu (D_T+1)(D_T+3)} \cdot r_{\max}^{D_T+3} - \frac{2\pi \tau_0 \cdot m^{\frac{3}{4}}}{3\mu} r_{\max}^3 \end{aligned} \quad (24)$$

The total flow through a cross-section of the core is:

$$\begin{aligned} Q &= \int_{r_{\min}^{\sqrt{m}}}^{r_{\max}^{\sqrt{m}}} q(r) dN(r) = \int_{r_{\min}^{\sqrt{m}}}^{r_{\max}^{\sqrt{m}}} q(r) dN(r) \\ &= \frac{\Delta P \cdot 2^{D_T-1} \cdot \pi \cdot L_0^{-D_T} \cdot m^{\frac{2D_T-D_f+6}{4}} \cdot (-D_f)}{\mu (D_T+1)(D_T+3)(D_T-D_f+3)} \left(r_{\max}^{D_T+3} - r_{\max}^{D_f} \cdot r_{\min}^{D_T-D_f+3} \right) \\ &\quad + \tau_0 \frac{2\pi \cdot m^{\frac{6-D_f}{4}} \cdot D_f}{3\mu(3-D_f)} (r_{\max}^3 - r_{\max}^{D_f} \cdot r_{\min}^{3-D_f}) \end{aligned} \quad (25)$$

Eq. (25) represents the nonlinear percolation mathematical model of high-PPT oil in the capillary bundle model. The expression of PTPG can be obtained at the condition of $Q = 0$:

$$\begin{aligned} G &= \frac{\Delta P}{L} \\ &= \tau_0 \frac{2^{2-D_T} \cdot m^{\frac{-D_T}{2}} (D_T+1)(D_T+3)(D_T-D_f+3)}{3(3-D_f)L_0^{1-D_T}} \cdot \frac{r_{\max}^{D_f} - r_{\max}^{D_f} \cdot \left(\frac{r_{\max}}{r_{\min}} \right)^{3-D_f}}{r_{\max}^{D_T+D_f} - r_{\max}^{D_f} \cdot r_{\min}^{D_T} \cdot \left(\frac{r_{\max}}{r_{\min}} \right)^{3-D_f}} \end{aligned} \quad (26)$$

In Eq. (26), $\left(\frac{r_{\max}}{r_{\min}} \right)^{3-D_f} \ll 1$, and can be neglected, simplifying the equation to:

$$G = \tau_0 \frac{2^{2-D_T} \cdot m^{\frac{-D_T}{2}} (D_T+1)(D_T+3)(D_T-D_f+3)}{3(3-D_f)L_0^{1-D_T}} \cdot \frac{1}{r_{\max}^{D_T}} \quad (27)$$

Therefore, Eq. (27) constitutes the PTPG model of high-PPT oil considering the pore throat structure of porous media, wax deposition damage, and fluid rheology. It is evident that G is positively correlated with τ_0 , D_f and D_T , and negatively correlated with $r_{\max}$ and m .

After deriving the prediction model, it is crucial to verify its accuracy. The pore throat structure parameters ($r_{\max}$, D_f , and D_T) can be obtained from the mercury injection data and fractal theory outlined in Sec. 2; the yield stress τ_0 can be obtained from rheological tests described in Sec. 4.1; and the deposition damage m can be determined from wax deposition experiments in Sec. 4.2. These parameters are incorporated into Eq. (27) to obtain predicted values of PTPG across various oil mobilities, which are then compared with experimental values, as illustrated in Fig. 14. The $G-\lambda_{oil}$ curve resulting from the model conforms well to a strong power-law form ($G = 5.106 \cdot \lambda_{oil}^{-1.048}$). The absolute error of the prediction model, calculated by determining the percentage of the area enclosed between the two curves relative to that enclosed by the experimental curve and the coordinate axis, is found to be 9.43 %, falling within an acceptable range. The slight underestimation of the predicted value compared to the experimental one can be attributed to the simplification of wax deposition in porous media during modeling. In practical wax precipitation scenarios, not all wax crystals deposit on pore surfaces to form deposition layers; some wax particles migrate with the oil flow and can block or completely plug pore throats, significantly increasing flow resistance. This phenomenon is not fully captured in the model, particularly when λ_{oil} is low. In such cases, the ratio of pore size to wax crystal size diminishes, exacerbating pore throat blockage and resulting in greater deviation between predicted and experimental results. Anyhow, from the practical application perspective, this model is a step toward the goal of accurately modeling the threshold pressure in high-PPT oil reservoirs.

6. Conclusions

Developing high-PPT oil reservoirs has emerged as a significant alternative to conventional ones, attracting increasing attention. However, improper development practices often result in wax crystallization, crude oil thickening, pore blockage, and other phenomena, ultimately deteriorating the percolation capacity of high-PPT oil. This, in turn, results in high natural decline rates for high-PPT oil reservoirs, rapid increases in water cut, low oil recovery rates, and in some cases, the abandonment of oilfields. This paper explores the internal factors contributing to the deterioration of high-PPT oil's percolation capacity, focusing on reservoir and crude oil characteristics. Through experiments and analytical methods, the non-Darcy percolation characteristics are comprehensively examined, yielding the following specific conclusions:

- (1) Significant variations in reservoir physical properties and complex pore structure deviate from Darcy's law in reservoir percolation behavior, potentially posing challenges related to the occurrence of threshold pressure characteristics.

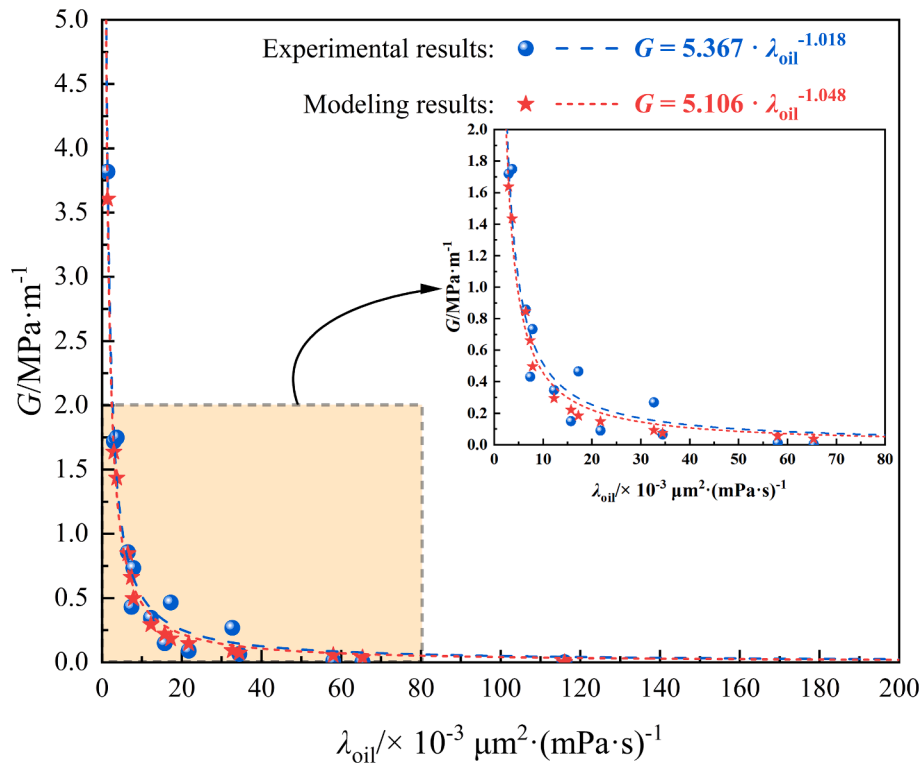


Fig. 14. Contrast diagram of pseudo threshold pressure gradient calculated from model with experimental results.

- (2) As the “cold damage” degree intensifies, the non-Newtonian rheological behavior of high-PPT oil becomes more pronounced. The $\eta-T$ relationship exhibits a unique trilinear rising pattern, and the yield stress also increases sharply, contributing to the deterioration of percolation capacity. A modified Arrhenius formula incorporating the structural rupture temperature T_0 and the non-Arrhenius index n accurately predicts viscosity under complex conditions.
- (3) Based on the capillary bundle model, a wax deposition model in porous media is established. Changes in pore radius, porosity before and after wax deposition, and wax deposition per unit volume are quantitatively analyzed under different temperature and permeability conditions.
- (4) The PTPG value increases with decreasing oil mobility, conforming to a remarkable power-law form ($G = 5.367 \cdot \lambda_{oil}^{-1.018}$). A flow boundary chart divides the flow state of high-PPT oil into flowable and non-flowable regions. In oilfield development, ensuring oil mobility exceeds the critical mobility is crucial, with the establishment of the ultimate drainage radius formula guiding well spacing design.
- (5) Considering the fractal characteristics of reservoir, non-Newtonian characteristics, and wax deposition damage, an analytical model of PTPG is developed, also conforming to a strong power-law form ($G = 5.106 \cdot \lambda_{oil}^{-1.048}$). The model’s accuracy is validated, with an absolute error between experimental and modeling curves of 9.43 %, falling within an acceptable range

The results here may advance the comprehension and prediction of high-PPT oil flow characteristics within porous media, offering valuable insights for efficient exploitation, gathering, and transportation in oil field operations.

CRediT authorship contribution statement

Xuechen Tang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yiqiang Li:** Validation, Supervision, Resources, Project administration, Funding acquisition. **Xinji Du:** Writing – original draft, Visualization, Conceptualization. **Chuanmin Xiao:** Resources, Project administration. **Zihao Li:** Investigation, Data curation. **Junqi Zeng:** Formal analysis, Data curation. **Jinxin Cao:** Visualization, Validation, Methodology. **Zheyu Liu:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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